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Artikel

The Effect of Screen Time on University Students' Sleep Quality: A Quantitative Study Using the Z-Test for the Difference of Two Means

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Abstract: This study aims to analyze the effect of *screen time* on the sleep quality of university students. Data were collected from 150 student respondents who reported their daily screen usage duration and sleep quality scores (scale 1–10). Respondents were grouped into two categories: high *screen time* (>6 hours/day, n=52) and non-high *screen time* (≤6 hours/day, n=98). Hypothesis testing was conducted using a Z-test for the difference of two means. The results show that the high *screen time* group had a mean sleep quality of 4.68, whereas the non-high group had a mean of 6.49. The obtained test statistic was $z = -7.96$ with a p-value ≈ 0.0000, far below the significance level $\alpha = 0.05$. Accordingly, the null hypothesis was rejected, and it is concluded that there is a significant difference in sleep quality between the two groups. The Pearson correlation coefficient between *screen time* and sleep quality is $r = -0.75$, indicating a strong negative relationship.

Keywords: *screen time*; sleep quality; Z-test; university students; hypothesis testing

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1. Introduction

In the current digital era, electronic devices such as *smartphones*, laptops, and tablets have become an inseparable part of university students' lives. According to a World Health Organization report [1], the use of digital screens among young people has increased significantly over the past decade. This phenomenon raises concerns about health impacts, particularly sleep quality, which is an important factor in academic performance and both physical and mental well-being.

Several previous studies have shown that blue light exposure from digital screens can suppress the production of the hormone melatonin, which plays a role in regulating the sleep–wake cycle or circadian rhythm [2]. Through a systematic review, Hale and Guan [3] found a consistent association between screen use and sleep disturbances in adolescents. However, research specifically addressing the impact of *screen time* on the sleep quality of university students in Indonesia remains limited.

Based on this background, the study poses the following research question: *Is there a significant difference in sleep quality between students with high screen time (>6 hours/day) and students with non-high screen time (≤6 hours/day)?* This question is framed in a neutral manner without assuming a particular direction, consistent with the principles of formulating a sound *research question*.

The hypotheses proposed in this study are as follows. The null hypothesis (H_0) states that there is no difference in mean sleep quality between the two groups, or formally $\mu_A - \mu_B = 0$. The alternative hypothesis (H_1) states that there is a difference in mean sleep quality between the two groups, namely $\mu_A - \mu_B \neq 0$. Testing is conducted using a two-tailed Z-test with a significance level of $\alpha = 0.05$.

The structure of this article is as follows. Section 2 discusses related work on the relationship between *screen time* and sleep quality. Section 3 describes the research methodology, including data collection and the statistical formulas used. Section 4 presents the experimental results and statistical analysis. Section 5 concludes the article.

2. Related Work

The relationship between the use of digital devices and sleep quality has become an active research topic in recent years. Carter et al. [4] conducted a meta-analysis of 20 studies and found that the use of portable screen-based devices is consistently associated with reductions in both the quantity and quality of sleep. The meta-analysis showed that active users of screen-based devices had twice the risk of experiencing poor sleep quality compared with non-users.

Exelmans and Van den Bulck [5] investigated pre-sleep *smartphone* use in an adult population in Belgium. That study found that *smartphone* use in bed correlated positively with longer sleep latency, poorer sleep quality, and greater daytime sleepiness. These findings are consistent with the study by Fossum et al. [6], which reported that the use of electronic media in bed is associated with insomnia symptoms.

Christensen et al. [7] used direct measurement of *smartphone screen time* and reported a mean daily usage duration of 3.7 hours in a young adult population. That study also revealed a negative correlation between usage duration and self-reported sleep quality. Demirci et al. [8] specifically examined university students and found a significant association between the severity of *smartphone* use, sleep quality, depression, and anxiety.

From the biological mechanism standpoint, Chang et al. [2] demonstrated that light exposure from an *e-reader* in the evening significantly suppresses melatonin production, shifts the circadian rhythm, and reduces morning alertness. Gradisar et al. [9], through a national survey in the United States, found that 95% of respondents used electronic devices within an hour before bedtime, which was associated with reduced sleep duration and quality.

Based on the literature review above, it is evident that the majority of studies have been conducted in Western countries. This study aims to fill that gap by analyzing the impact of *screen time* on the sleep quality of university students using a statistical hypothesis testing approach, specifically the Z-test for the difference of two means.

3. Methodology

3.1. Research Flow

The methodology of this study follows a systematic flow, starting from problem identification and ending with the drawing of conclusions. Figure 1 illustrates the research flow diagram used. Each stage is carried out sequentially to ensure the validity of the research results.

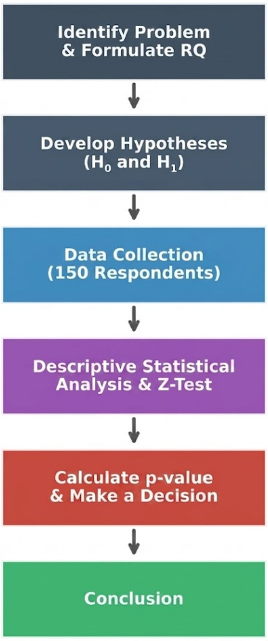

Figure 1. Research methodology flow diagram.

3.2. Data Collection

Data were collected from 150 university students through a questionnaire that included demographic information, daily *screen time* duration (in hours), and self-reported sleep quality scores on a scale of 1–10. Respondents consisted of 68 males (45.3%) and 82 females (54.7%). Based on *screen time* duration, respondents were grouped into three categories: high (>6 hours, $n = 52$), medium (3–6 hours, $n = 79$), and low ($\leq$3 hours, $n = 19$).

3.3. Statistical Formulas

To test the research hypothesis, a Z-test for the difference of two independent means was used. This method was chosen because the sample sizes of both groups are sufficiently large ($n > 30$), so that the sampling distribution approaches the normal distribution according to the Central Limit Theorem [10]. The calculation steps are as follows.

First, for each group $k \in \{A, B\}$, the sample mean and standard deviation are computed:

$$\bar{x}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} x_{k,i}, \quad \sigma_k = \sqrt{\frac{1}{N_k} \sum_{i=1}^{N_k} (x_{k,i} - \bar{x}_k)^2} \tag{1}$$

where N_k is the sample size of group k and $x_{k,i}$ is the sleep quality value of the i-th respondent in that group.

Next, the *Standard Error of the Mean* (SEM) and the *Standard Error of the Difference* are computed as:

$$\text{SEM}_k = \frac{\sigma_k}{\sqrt{N_k}}, \quad SE_{A-B} = \sqrt{\text{SEM}_A^2 + \text{SEM}_B^2} \tag{2}$$

The Z test statistic is then computed using:

$$z = \frac{\Delta\bar{x} - 0}{SE_{A-B}} = \frac{(\bar{x}_A - \bar{x}_B)}{SE_{A-B}} \tag{3}$$

Finally, the p-value is obtained from the standard normal distribution table (Z-table). Since the test is two-tailed, p-value $= 2 \times P(Z \leq z)$ for $z < 0$. The decision is made by comparing the p-value to the significance level $\alpha = 0.05$: if p-value $< \alpha$, then H_0 is rejected.

4. Experimental Results

4.1. Descriptive Statistics

Table 1 presents a summary of descriptive statistics for each *screen time* category. Overall, the mean *screen time* across all respondents is 5.34 hours/day with a mean sleep quality of 5.86. A clear pattern emerges: the higher the *screen time*, the lower the sleep quality score.

Table 1. Descriptive statistics by *screen time* category.

Category	n	Mean ST (hours)	Mean SQ (score)
Low ($\leq$3 hours)	19	2.31	7.39
Medium (3–6 hours)	79	4.77	6.28
High (>6 hours)	52	7.31	4.68
Total	**150**	**5.34**	**5.86**

The low *screen time* group has the highest sleep quality score (7.39), while the high group has the lowest score (4.68). The difference in sleep quality between the low and high groups reaches 2.71 points, indicating a substantial difference at the descriptive level.

Figure 2 displays a scatter plot of the relationship between *screen time* and sleep quality for all 150 respondents. The observed pattern shows a downward trend: the higher the *screen time* duration, the lower the reported sleep quality. The Pearson correlation coefficient obtained is $r = -0.75$, indicating a strong negative relationship between the two variables.

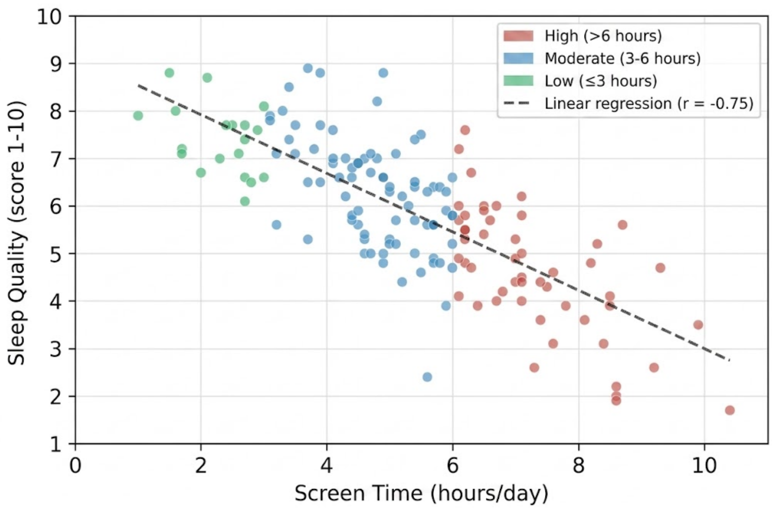

Figure 2. Scatter plot of *screen time* vs. sleep quality ($r = -0.75$).

4.2. Hypothesis Testing

Hypothesis testing was conducted to compare the mean sleep quality between the high *screen time* group (Group A, $n_A = 52$) and the non-high group (Group B, $n_B = 98$). Table 2 presents the statistical calculation results for both groups.

Table 2. Statistical calculation results for the Z-test.

Statistic	Group A (High ST)	Group B (Non-High ST)
Sample size (n)	52	98
Mean ($\bar{x}$)	4.6769	6.4939
Standard deviation (σ)	1.3990	1.1899
SEM ($\sigma/\sqrt{n}$)	0.1940	0.1202

Based on the data in Table 2, calculations are performed using Equations (2) and (3) as follows:

- **Standard Error of the Difference:** $SE_{A-B} = \sqrt{0.1940^2 + 0.1202^2} = \sqrt{0.037638 + 0.014448} = \mathbf{0.2282}$
- **Difference between sample means:** $\Delta\bar{x} = \bar{x}_A - \bar{x}_B = 4.6769 - 6.4939 = \mathbf{-1.8170}$
- **Test statistic:** $z = \frac{-1.8170-0}{0.2282} = \mathbf{-7.9613}$
- **p-value:** Since $|z| = 7.96$ far exceeds the Z-table limit (at $z = -3.6$, $P = 0.0002$), then p-value $= 2 \times P(Z \leq -7.96) \approx 2 \times 0.0000 = \mathbf{0.0000}$

4.3. Decision and Interpretation

Since p-value $\approx 0.0000 < \alpha = 0.05$, the **null hypothesis (H_0) is rejected**. The alternative hypothesis (H_1) is therefore accepted: there is a statistically significant difference in mean sleep quality between university students with high and non-high *screen time*.

Figure 3 visualizes the distribution of sleep quality across the three *screen time* categories. A consistent downward pattern is clearly visible from the low group to the high group, supporting the results of the hypothesis test that was conducted.

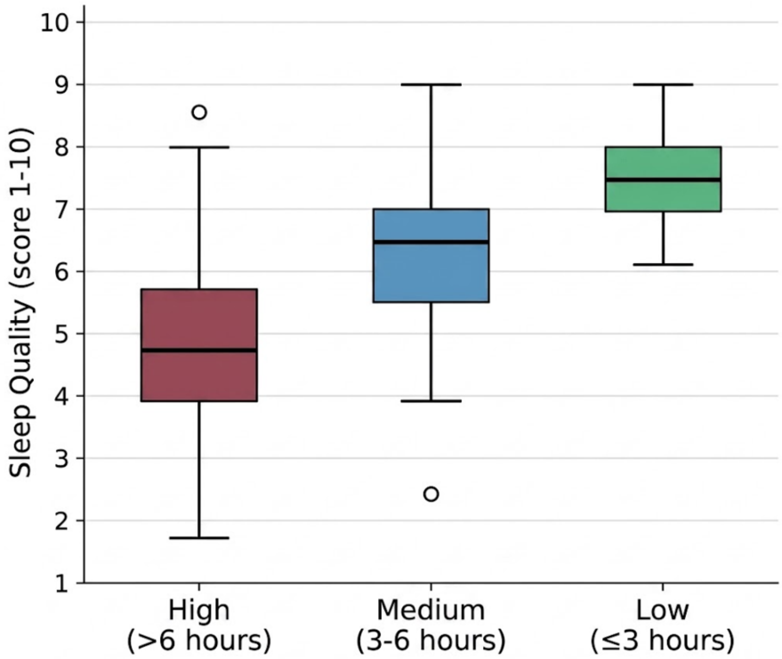

Figure 3. Distribution of sleep quality by *screen time* category.

In practical terms, the 1.82-point difference (on a 1–10 scale) between the high and non-high *screen time* groups represents a fairly large gap. The correlation value of $r = -0.75$ also indicates that *screen time* can account for approximately 56% of the variation in sleep quality ($r^2 = 0.56$). These findings are consistent with previous literature showing the negative impact of excessive screen use on sleep quality.

5. Conclusions

This study analyzes the effect of *screen time* on the sleep quality of university students using data from 150 respondents and a Z-test for the difference of two means. The testing yields $z = -7.96$ with a p-value ≈ 0.0000, so the null hypothesis is rejected at a significance level of $\alpha = 0.05$. There is a statistically significant difference between the mean sleep quality of students with high *screen time* ($\bar{x} = 4.68$) and non-high *screen time* ($\bar{x} = 6.49$).

The Pearson correlation coefficient of $r = -0.75$ confirms a strong negative relationship between screen usage duration and sleep quality. These findings are in line with previous studies [2–4] and provide empirical evidence in the context of university students in

Indonesia. A practical implication of this study is the importance of *screen time* management for university students in order to maintain optimal sleep quality.

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Conflict of Interest: The author declares no conflict of interest.

The Use of AI: The development of paragraphs in the Related Work and Introduction sections was assisted by AI. All core ideas, data analysis, statistical computations, and interpretations of results are the author's own work.

References

1. World Health Organization. Screen time and digital wellness among young adults. *WHO Technical Report* **2024**.
2. Chang, A.M.; Aeschbach, D.; Duffy, J.F.; Czeisler, C.A. Evening use of light-emitting eReaders negatively affects sleep, circadian timing, and next-morning alertness. *Proceedings of the National Academy of Sciences* **2015**, *112*, 1232–1237.
3. Hale, L.; Guan, S. Screen time and sleep among school-aged children and adolescents: a systematic literature review. *Sleep Medicine Reviews* **2015**, *21*, 50–58.
4. Carter, B.; Rees, P.; Hale, L.; Bhattacharjee, D.; Paradkar, M.S. Association between portable screen-based media device access or use and sleep outcomes: a systematic review and meta-analysis. *JAMA Pediatrics* **2016**, *170*, 1202–1208.
5. Exelmans, L.; Van den Bulck, J. Bedtime mobile phone use and sleep in adults. *Social Science & Medicine* **2016**, *148*, 93–101.
6. Fossum, I.N.; Nordnes, L.T.; Storemark, S.S.; Bjorvatn, B.; Pallesen, S. The association between use of electronic media in bed before going to sleep and insomnia symptoms, daytime sleepiness, morningness, and chronotype. *Behavioral Sleep Medicine* **2014**, *12*, 343–357.
7. Christensen, M.A.; Bettencourt, L.; Kaye, L.; Moturu, S.T.; Nguyen, K.T.; Olgin, J.E.; Pletcher, M.J.; Marcus, G.M. Direct measurements of smartphone screen-time: relationships with demographics and sleep. *PloS One* **2016**, *11*, e0165331.
8. Demirci, K.; Akgönül, M.; Akpinar, A. Relationship of smartphone use severity with sleep quality, depression, and anxiety in university students. *Journal of Behavioral Addictions* **2015**, *4*, 85–92.
9. Gradisar, M.; Wolfson, A.R.; Harvey, A.G.; Hale, L.; Rosenberg, R.; Czeisler, C.A. The sleep and technology use of Americans: findings from the National Sleep Foundation's 2011 Sleep in America poll. *Journal of Clinical Sleep Medicine* **2013**, *9*, 1291–1299.
10. Moore, D.S.; Notz, W.I.; Fligner, M.A. *The Basic Practice of Statistics*, 7th ed.; W. H. Freeman, 2016.